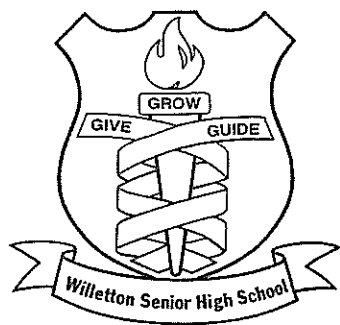




Semester One Examination, 2013



CHEMISTRY 3AB



See next page

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Question 26

(11 marks)

(a) Write the electron configuration for each of the species below.

(2 marks)

<i>Species</i>	<i>Electron configuration</i>
Sulfide ion	2 8 8
Aluminium atom	2 8 3

(b) Complete the table below

All valence electron pairs should be presented either as : or —

(9 marks)

<i>Species</i>	<i>Lewis electron dot diagram</i>	<i>Sketch or name of shape</i>	<i>Polarity of molecule (polar or non-polar)</i>
NBr ₃		Pyramidal	Polar
OBr ₂		Bent	Polar
SO ₃		Triangular Planar	Non-polar

Question 27
(5 marks)

- (a) Place each of the following substances in the appropriate column of the table below on the basis of the experimental observation stated in the table below. (2 marks)



<i>Conducts electricity well but does not dissolve in dilute hydrochloric acid</i>	<i>Boils at a temperature below 100°C</i>	<i>Forms a pale green solution when added to a dilute solution of ethanoic acid</i>	<i>Can only conduct electricity in molten/aqueous solution</i>
Cu_(s)	Br_{2(l)}	Fe_(s)	NH₄NO_{2(s)}

- (b) State the most significant type of intermolecular forces for each of the following liquids. (3 marks)

Liquids	Most significant intermolecular force
Sulfur (S ₈)	Dispersion forces
CH ₃ F	Dipole-dipole
H ₂ SO ₄	Hydrogen bonds

Question 28
(3 marks)

Write ONE relevant ionic chemical equation to explain each of the observations shown in the table below.

Observation	Explanatory equation
The pH of a solution of NaHCO ₃ is greater than 7.	$\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{OH}^- + \text{H}_2\text{CO}_3$
The addition of solid NaNH ₂ to water produces a pungent odour and turns red litmus paper to blue.	$\text{NH}_2^- + \text{H}_2\text{O} \rightleftharpoons \text{OH}^- + \text{NH}_3$
A vinegar smell is produced when a colourless liquid is added to a dilute solution of potassium ethanoate.	$\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$

Question 29**(5 marks)**

Explain the following using relevant concepts and theories.

- (a) The first ionization energy of a fluorine atom is higher than that of a boron atom. (3 marks)

Both have the same number of electron shells i.e. shielding effect not a factor

F has a larger number of protons than B i.e. a larger nuclear charge (1)

F can exert a larger pull on its outermost electron than B can (1)

More energy is thus needed to remove outermost electron from F compared to B (1)

- (b) The radius of a fluoride ion (F^-) is larger than the radius of sodium ion (Na^+) even though both have the same number of electrons. (2 marks)

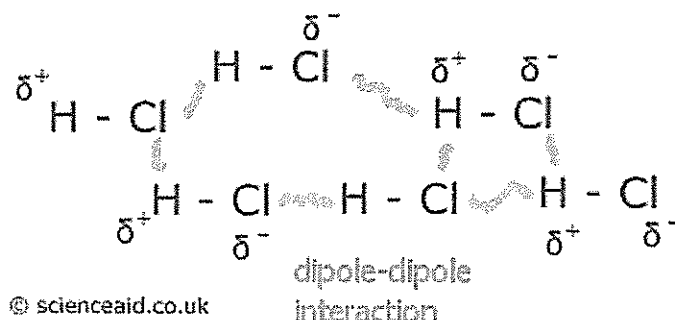
Higher number of protons per electron (or larger proton : electron ratio) for Na^+ compared to F^- (1)

i.e. valence electrons in Na^+ experience a stronger pull towards nucleus compared to F^- and are thus closer to the nucleus hence a smaller radius (1)

Question 30

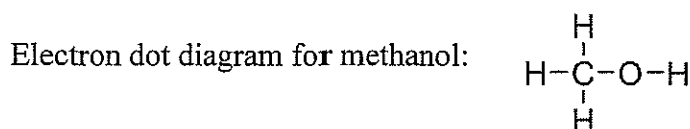
(6 marks)

The following diagram shows dipole-dipole interaction in liquid hydrogen chloride.



Liquid methanol (CH_3OH) and solid sodium hydroxide (NaOH) are both soluble in water.

- (a) Use the diagram above as a guide to draw relevant diagrams showing the most dominant form of interaction between EACH of the two substances above and liquid water molecules in solution. (Note: an electron dot diagram for methanol has been provided for your reference) (2 marks)



<i>methanol in water</i> (1)	<i>sodium hydroxide in water</i> (1)
Name of interaction: Hydrogen bonds (1)	Name of interaction: Ion-dipole (1)

- (b) Name the interaction that you have drawn in part (a) in the space provided above. (2 marks)

- (c) At atmospheric pressure, methanol has a boiling point of about 65°C while sodium hydroxide has a boiling point of approximately 1390°C . Briefly explain this difference using relevant concepts and theories.

(2 marks)

Methanol = covalent molecular substance having weak hydrogen bonds between liquid molecules i.e. little energy is needed to overcome the weak attraction hence a relatively low boiling point. (1)

Sodium hydroxide = ionic substance having strong electrostatic attraction between ions i.e. large amount of energy is needed to overcome the relatively stronger attraction. (1)

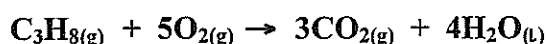
Question 31

(6 marks)

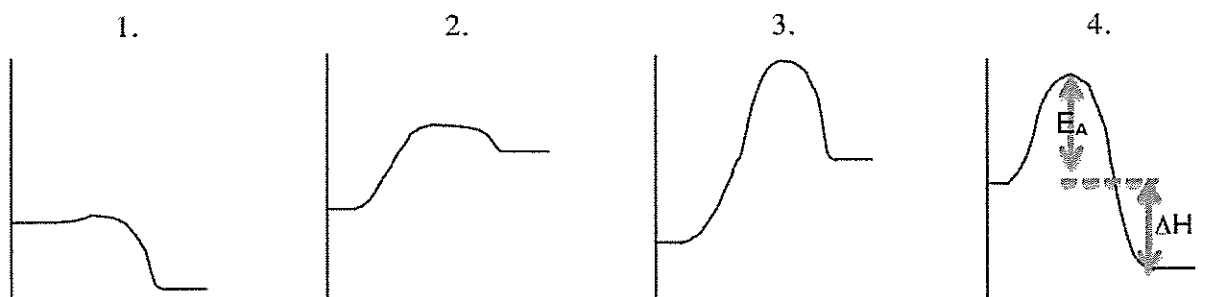
Propane is a hydrocarbon (C_3H_8) and is sometimes referred to as liquefied petroleum gas, LP-gas, or LPG. It is used as an alternative fuel to petroleum and diesel.

People use propane in and around their homes for water heaters, outdoor grills, fireplaces, and appliances. Industrial uses include propane-driven forklifts and fleet vehicles.

The chemical equation for the combustion of propane in air is



(a) Consider the four graphs below showing changes in energy (kJ mol^{-1}) against reaction coordinate



One of these graphs represents the combustion of propane. Identify this graph, giving reasons for your choice. Use lines or markings to clearly show activation energy and heat of reaction on your chosen graph. (4 marks)

Graph 4 (1)

Significant amount of energy is needed to start the reaction (flame/match ...) hence the high E_A (1)

Large amount of heat released to the surrounding by reaction – hence the exothermic shape. (1)

Labeling of E_A and ΔH on graph (1)

(b) Identify the graph that represents a process **most likely** to proceed spontaneously at room temperature. Briefly explain your choice. (2 marks)

Graph 1 (1)

Reason: very little amount of E_A is required for the reaction (i.e. can be had from environmental heat) (1)

Question 32

(6 marks)

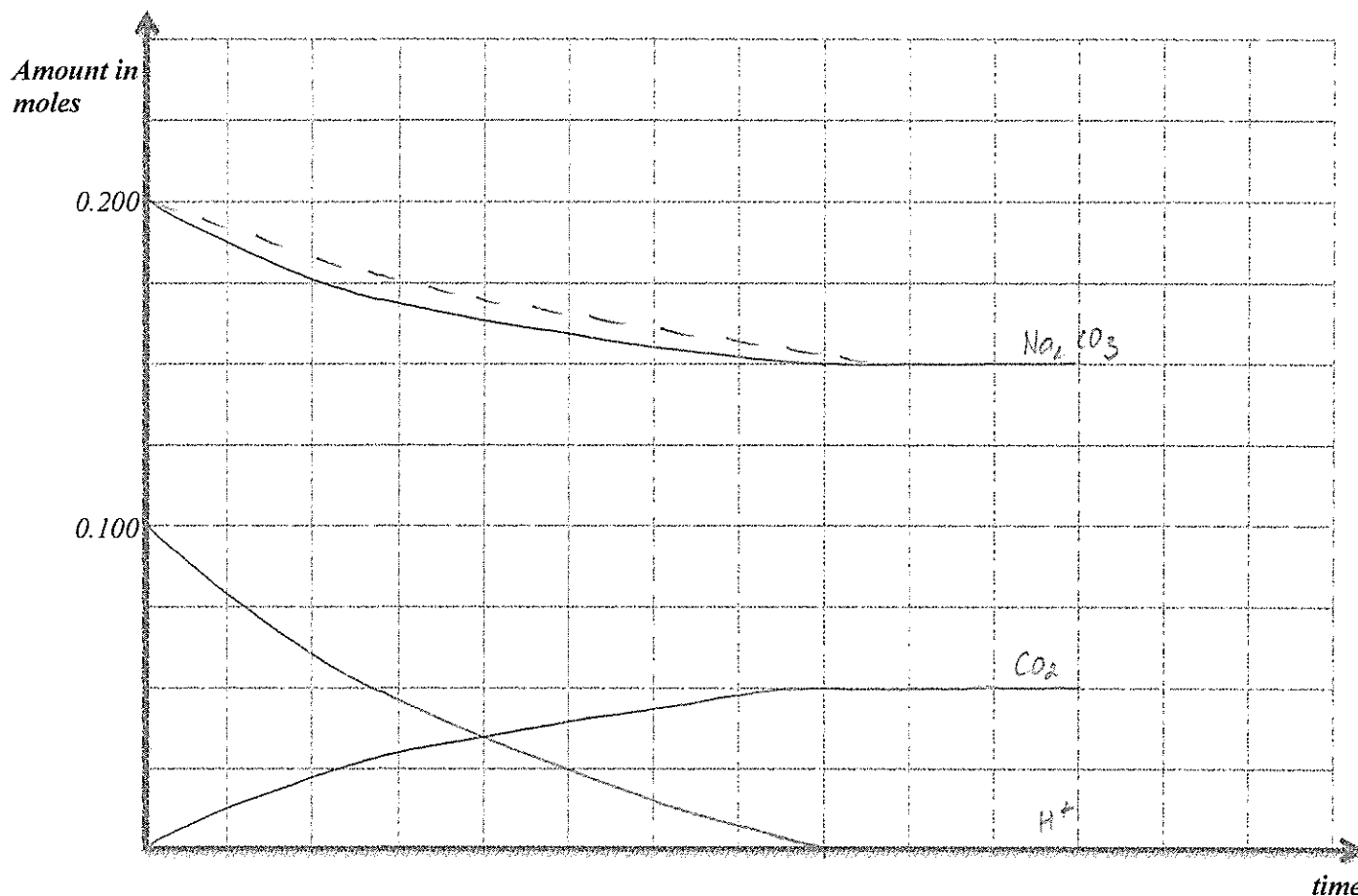
Consider the following reaction:



In an experiment 0.200 mole of sodium carbonate powder and 0.100 mole of hydrochloric acid were placed in an open beaker.

The reaction was allowed to proceed to completion.

- (a) On the axes below sketch graphs to illustrate the rate of disappearance of the sodium carbonate powder and the hydrochloric acid over time. Also sketch the rate of appearance of the carbon dioxide. Clearly label **all three** graphs. (3 marks)



- (b) On the axes above add a new graph for $\text{Na}_2\text{CO}_{3(s)}$ (note: use a dotted line ----) showing the effect of using large crystals of sodium carbonate instead of the powder. Explain in terms of the Collision theory why this graph is the shape you have drawn. (3 marks)

Correctly drawn dotted line (1)

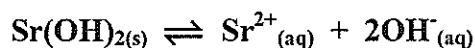
Large crystals means a small surface area available for reaction (1)

Number of collisions between reactants decrease as a result hence fewer collisions with sufficient E_A to form products i.e. rate decreases (1)

Question 33

(9 marks)

Strontium hydroxide, $\text{Sr}(\text{OH})_2$ is slightly soluble in water. The equilibrium established when solid $\text{Sr}(\text{OH})_2$ is in contact with its saturated solution is represented by



An equilibrium mixture of solid strontium hydroxide and its saturated solution is prepared. Four test tubes are set up, each containing some of the equilibrium mixture – a little white solid under a colourless solution. Each of the test tubes is treated as described below.

Complete the table below by indicating the direction of the expected shift in equilibrium following the changes stated in the table. Use Le Chatelier's theory to briefly explain how the shift was predicted.

<i>Test tube</i>	<i>Change</i>	<i>Direction of shift in equilibrium (write 'left', 'right' or no change)</i>	<i>Reasons for shift (or no change)</i>
A	A little solid strontium iodide is added and the mixture shaken	Left (1)	Adding SrI_2 increases $[\text{Sr}^{2+}]$ (1/2) System favours reverse reaction to counteract change (1/2)
B	A little concentrated hydrochloric acid is added and the mixture shaken	Right (1)	Adding HCl decreases $[\text{OH}^-]$ (1/2) due to $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}_{(l)}$ (1/2) System favours forward reaction to counteract change (1/2)
C	A little solid sodium carbonate is added and the mixture shaken	Right (1)	Adding Na_2CO_3 decreases $[\text{Sr}^{2+}]$ (1/2) due to $\text{Sr}^{+2} + \text{CO}_3^{-2} \rightarrow \text{SrCO}_{3(s)}$ (1/2) System favours forward reaction to counteract change (1/2)

Question 34
(7 marks)

Consider the following process at equilibrium.



- (a) Predict whether the following changes will increase, decrease or have no effect to the rate at which equilibrium is attained. (3 marks)

<i>Change</i>	<i>Effect</i>
Liquid water is added	No change (1)
The pressure of CH ₄ is increased	Increase (1)
The temperature is decreased to a new constant value	Decrease (1)

- (b) Predict whether the following changes will increase, decrease or have no effect on the equilibrium yield of the reaction and the value of equilibrium constant (K) (4 marks)

<i>Change</i>	<i>Effect on yield</i>	<i>Effect on equilibrium constant (K)</i>
The system is cooled to a new temperature	Decrease (1)	Decrease (1)
Decreasing the volume of the system	Decrease (1)	No change (1)

Question 35
(6 marks)

Complete the following table below by giving a brief description of a **chemical** test that could be used to distinguish between the pair of substances listed. State relevant observations relating to the test used for each of substance 1 and substance 2.

<i>Substances to be distinguished</i>		<i>Brief description of chemical test (chemical equations are <u>not</u> required)</i>	<i>Observation (write nvr if no visible change is observed)</i>	
<i>Substance 1</i>	<i>Substance 2</i>		<i>Substance 1</i>	<i>Substance 2</i>
0.5 mol/L sodium sulfate solution	0.5 mol/L sodium hydroxide solution	Add a solution containing Ba ²⁺ (1)	White ppt (1)	Nvr (1)
solid iron (II) carbonate	solid iron (II) hydroxide	Add dilute H ⁺ (1)	Green solid dissolves Colourless gas forms Solution turns green (1)	Green solid dissolves No gas forms Solution turns green (1)

See next page

Question 36

(6 marks)

Write **balanced ionic equations** and observations for each of the following. In each case describe in full what you would observe, including

- any colours
- odours
- precipitates (give colour)
- gases evolved (give colour or describe as colourless).

If no change is observed, you should state this.

- (a) 20.0 mL of 0.5 molL⁻¹ calcium chloride solution is mixed with 100.0 mL of 1.0 molL⁻¹ lead (II) nitrate solution. (3 marks)

<i>Equation</i>	$2 \text{Cl}^-_{(\text{aq})} + \text{Pb}^{2+}_{(\text{aq})} \rightarrow \text{PbCl}_{2(\text{s})}$	(2)
<i>Observation</i>	White precipitate forms, solution stays colourless	(1)

- (b) Solid calcium hydroxide is added to dilute ammonium iodide solution. (3 marks)

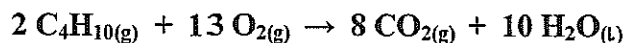
<i>Equation</i>	$\text{Ca(OH)}_{2(\text{s})} + 2 \text{NH}_4^+_{(\text{aq})} \rightarrow 2 \text{NH}_{3(\text{aq})} + 2 \text{H}_2\text{O}_{(\text{l})} + \text{Ca}^{2+}_{(\text{aq})}$	(2)
<i>Observation</i>	White solid dissolves, pungent odour forms, solution stays colourless	(1)

Question 37

(14 marks)

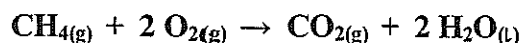
A heater company produces a portable gas heater that can run on either butane gas (C_4H_{10}) cylinders or household supply of natural gas (CH_4).

- i. The equation for the complete combustion of butane gas is as follows:



In normal use, one gas cylinder containing 500.0g of butane will give exactly 150 hours of use.

- ii. The equation for the complete combustion of natural gas is as follows:



The methane from the household supply system is delivered at 200.0 kPa pressure and at a rate of 2.40 L per hour.

Assume a temperature of 20.0°C throughout.

- (a) Which gas (butane or methane) requires the most mass (in grams) of oxygen **per hour** to sustain complete combustion of the butane during normal operation? (show all working) (7 marks)

$$n(\text{C}_4\text{H}_{10}) = 500 / 58.12 = 8.603 \text{ mol} \quad (1/2)$$

$$n(\text{O}_2) \text{ required} = 13/2 \times 8.603 = 55.919 \text{ mol} \quad (1)$$

$$n(\text{O}_2 \text{ per hour}) = 55.919 / 150 = 0.373 \text{ mol} \quad (1)$$

$$m(\text{O}_2 \text{ per hour}) = 0.373 \times 32 = 11.9 \text{ g} \quad (1)$$

$$n(\text{CH}_4) = 200 \times 2.40 / 8.315 \times 293 = 0.197 \text{ mol} \quad (1/2)$$

$$n(\text{O}_2) \text{ required} = 2/1 \times 0.197 = 0.394 \text{ mol} \quad (1)$$

$$m(\text{O}_2 \text{ per hour}) = 0.394 \times 32 = 12.6 \text{ g} \quad (1)$$

$$\text{CH}_4 \text{ requires the most mass of O}_2 \text{ per hour} \quad (1)$$

- (b) Calculate the mass of water produced during 10.0 hours of normal operation when running on the butane gas cylinders. (2 marks)

$$n(\text{H}_2\text{O}) = 10/2 \times 8.603 = 43.014 \text{ mol} \quad (1/2)$$

$$n(\text{H}_2\text{O}) \text{ in 10 hours} = (43.014 / 150) \times 10 = 2.868 \text{ mol} \quad (1)$$

$$m(\text{H}_2\text{O}) = 2.868 \times 18.016 = 51.7 \text{ g} \quad (1/2)$$

On the first of July 2012, the Australian government imposed a carbon tax level of \$23.0 per tonne of carbon dioxide emitted on a number of manufacturing industries (1 tonne = 1×10^6 g).

Like many other industries, this heater company immediately announced a price increase of \$5.00 on this portable heater model to help it 'pay for the new tax'.

- (c) Assuming the heater is operating from a household methane gas system and is used for 100 days a year, calculate the total mass of carbon dioxide produced by this portable heater over a period of 10 years (life expectancy of a heater) and state whether this price increase is fair for the consumers. (5 marks)

$$n(\text{CO}_2) = n(\text{CH}_4) = 0.197 \text{ mol per hour}$$

$$n(\text{CO}_2) \text{ over 100 days \& 10 years} = 0.197 \times 24 \times 100 \times 10 = 4728.49 \text{ mol} \quad (1)$$

$$m(\text{CO}_2) = 4728.49 \times 44.01 = 208100.9 \text{ g} \quad (1)$$

$$\text{Carbon tax per g} = 23 / 1 \times 10^6 = \$ 2.3 \times 10^{-5} \quad (1)$$

$$\text{Carbon tax of heater} = \$ 2.3 \times 10^{-5} \times 208100.9 \text{g} = \$ 4.79 \quad (1)$$

i.e. The increase is reasonably fair. (1)

(Note: Some students may state that an increase of \$0.20 is not reasonable ☺)

Question 38

(13 marks)

The preparation of sulfuric acid from iron pyrites (FeS_2) can be represented by the following series of reactions:



- (a) How many moles of sulfuric acid would be produced from **ONE** mole of iron pyrites? (1 mark)

2 moles (1)

- (b) What mass of iron pyrites would be needed to produce 123 tonne ($1 \text{ tonne} = 1 \times 10^6 \text{ g}$) of sulfuric acid if the Contact process only converts 92.5% of the sulfur dioxide and the other reactions achieve 100% conversion? (6 marks)

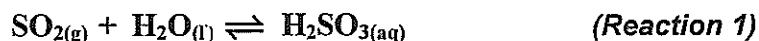
$$n(\text{H}_2\text{SO}_4) = 123000000 / 98.076 = 1.254 \times 10^6 \text{ mol} \quad (2)$$

$$n(\text{FeS}_2) \text{ assuming } 100\% \text{ efficiency} = 1.254 \times 10^6 / 2 = 6.27 \times 10^5 \text{ mol} \quad (1)$$

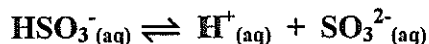
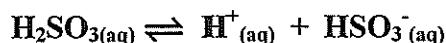
$$n(\text{FeS}_2) \text{ required for } 92.5\% \text{ efficiency} = 100 / 92.5 \times 6.27 \times 10^5 = 6.78 \times 10^5 \text{ mol} \quad (2)$$

$$m(\text{FeS}_2) = 6.78 \times 10^5 \times 119.97 = 8.13 \times 10^7 \text{ g} \quad (1)$$

Sulfurous acid (H_2SO_3) can be produced by the dissolving of sulfur dioxide gas in liquid water under the right conditions.



Sulfurous acid partially ionises in solution



- (c) If **Reaction 1** above is slightly exothermic then predict the effect of the following changes on the pH of the solution: (3 marks)

<i>Change to equilibrium</i>	<i>Effect on pH (write increase, decrease or no change)</i>
The temperature of the system is increased to a new constant value	Increase
Finely powdered $\text{CaCO}_{3(s)}$ is added	Increase
A little concentrated nitric acid is added	Decrease

- (d) The equilibrium constant (K) value for the ionisation of a 1.0 mol L^{-1} sulfurous acid solution is 1.40×10^{-2} at 25°C and atmospheric pressure and the equilibrium constant value for the ionisation of a 1.0 mol L^{-1} hydrogensulfite (HSO_3^-) solution is 6.30×10^{-8} at the same conditions.

Which of these two solutions (sulfurous acid or hydrogen sulfite) would be a stronger electrolyte at 25°C ? Explain your answer. (3 marks)

1 mol/L of H_2SO_3 can only produce $\ll 1 \text{ mol/L}$ of ions (H^+ and HSO_3^-) in total (small K value) (1)

1 mol/L of HSO_3^- already has 1 mol/L of ions (HSO_3^-) and an insignificant concentration of H^+ and SO_3^{2-} ions (very small K value) (1)

as total [ions] in $\text{HSO}_3^- > [\text{H}^+] \text{ and } [\text{SO}_3^{2-}]$ in H_2SO_3 it is thus a better electrolyte. (1)

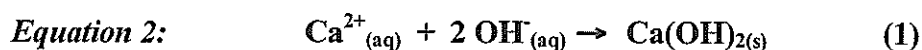
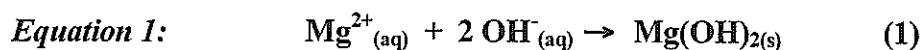
Question 39

(14 marks)

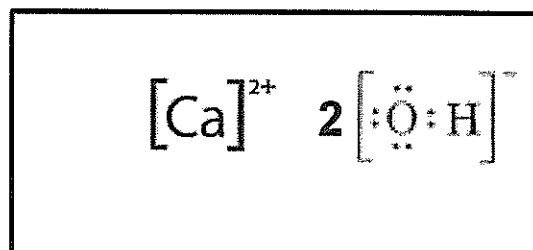
A local water sample was analysed and found to contain a high concentration of magnesium and calcium ions.

Common identification tests for the presence of these ions include the precipitation reaction of these ions with sodium hydroxide or sodium carbonate solutions.

- (a) Write relevant ionic equations to show the precipitation of magnesium and calcium ions from the water by dilute sodium hydroxide solution. (2 marks)



- (b) Draw a Lewis electron dot diagram for $\text{Ca}(\text{OH})_2$



(1 mark)

(1)

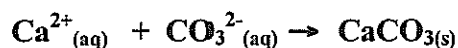
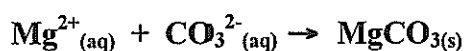
- (c) A 5.00 mL water sample has a magnesium ion concentration of 12.1 mgL^{-1} **and** a calcium ion concentration of 5.01 mgL^{-1} . Calculate the mass of calcium ions and magnesium ions in the water sample. (2 marks)

$$m(\text{Mg}^{2+}) = 12.1 \times 5/1000 = 0.0605 \text{ mg} \quad \text{or} \quad 6.05 \times 10^{-5} \text{ g} \quad (1)$$

$$m(\text{Ca}^{2+}) = 5.01 \times 5/1000 = 0.0250 \text{ mg} \quad \text{or} \quad 2.50 \times 10^{-5} \text{ g} \quad (1)$$

A scientist attempts to remove all magnesium and calcium ions from a polluted lake using precipitation reaction. In a small trial, he adds 50.0 g of sodium carbonate decahydrate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) to 10.0 L of polluted water.

The relevant chemical equations for the precipitation reactions are as follow:



The polluted water from the lake was found to contain 2.50 g of calcium ions and 1.50 g of magnesium ions.

- (d) Calculate the amount of carbonate ions required to remove all the calcium and magnesium ions from the water and hence determine whether the use of 50.0 g sodium carbonate decahydrate is enough for the total removal of these ions. (6 marks)

$$n(\text{Ca}^{2+}) = 2.5 / 40.1 = 0.0623 \text{ mol} \quad (1)$$

$$n(\text{Mg}^{2+}) = 1.5 / 24.3 = 0.0617 \text{ mol} \quad (1)$$

$$n(\text{CO}_3^{2-}) = n(\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}) = 50 / 286.15 = 0.175 \text{ mol} \quad (1)$$

$$n(\text{Ca}^{2+} \text{ \& \; } \text{Mg}^{2+}) \text{ total} = 0.124 \text{ mol} \quad (1)$$

$$n(\text{CO}_3^{2-}) > n(\text{Ca}^{2+} \text{ \& \; } \text{Mg}^{2+}) \text{ total} \quad (1)$$

$$\text{i.e. there is sufficient Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O for the job!} \quad (1)$$

- (e) From your answer to part (d),

- if the amount of sodium carbonate decahydrate is not enough, determine the mass of additional sodium carbonate decahydrate that needs to be added.

OR

- if the amount of sodium carbonate decahydrate is in excess, determine the mass of unreacted sodium carbonate decahydrate. (2 marks)

$$n(\text{CO}_3^{2-}) \text{ excess} = 0.175 - 0.124 = 0.0510 \text{ mol} \quad (1)$$

$$n(\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}) \text{ excess} = 0.0510 \times 286.15 = 14.5 \text{ g} \quad (1)$$

Question 40**(15 marks)**

Both perchloric acid, HClO_4 and barium hydroxide, $\text{Ba}(\text{OH})_2$ are substances that fully ionise/dissociate when in water solution.

- (a) Calculate the pH of a 50.0 mL solution of $0.0250 \text{ mol L}^{-1}$ perchloric acid. (1 mark)

$$\text{pH} = -\log(0.025) = 1.60 \quad (1)$$

- (b) Calculate the pH of a $0.0250 \text{ mol L}^{-1}$ solution of barium hydroxide. (3 marks)

$$[\text{OH}^-] = 2 \times [\text{Ba}(\text{OH})_2] = 0.05 \text{ mol/L} \quad (1)$$

$$[\text{H}^+] = 1 \times 10^{-14} / 0.05 = 2 \times 10^{-13} \quad (1)$$

$$\text{pH} = -\log(2 \times 10^{-13}) = 12.7 \quad (1)$$

- (c) In an experiment, a student dissolves 0.215 g of solid $\text{Ba}(\text{OH})_2$ in 50.0 mL of 0.025 mol L^{-1} perchloric acid. Calculate the final pH of the mixture. (6 marks)

$$n(\text{Ba}(\text{OH})_2) = 0.215 / 171.316 = 1.255 \times 10^{-3} \text{ mol} \quad (1/2)$$

$$n(\text{OH}^-) = 2.51 \times 10^{-3} \text{ mol} \quad (1/2)$$

$$n(\text{HClO}_4) = 50 / 1000 \times 0.025 = 1.25 \times 10^{-3} \text{ mol} \quad (1/2)$$

$$n(\text{H}^+) = 1.25 \times 10^{-3} \text{ mol} \quad (1/2)$$

$$n(\text{OH}^-) > n(\text{H}^+) \text{ i.e. } \text{OH}^- \text{ is in excess}$$

$$n(\text{OH}^-)_{\text{excess}} = 1.26 \times 10^{-3} \text{ mol} \quad (1)$$

$$[\text{OH}^-]_{\text{excess}} = 1.26 \times 10^{-3} / 0.05 = 2.5 \times 10^{-2} \text{ mol} \quad (1)$$

$$[\text{H}^+] = 1 \times 10^{-14} / 2.5 \times 10^{-2} = 3.97 \times 10^{-13} \quad (1)$$

$$\text{pH} = -\log(3.97 \times 10^{-13}) = 12.4 \quad (1)$$

- (d) Assuming the mass of 1.00 L of the final solution mixture is 1.02 kg, determine the concentration of barium ions (in **parts per million**) at the end of the reaction in part (c). (2 marks)

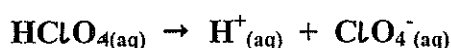
$$[\text{Ba}^{2+}] = n(\text{Ba}(\text{OH})_2) / V = 1.255 \times 10^{-3} / 0.05 = 0.0251 \text{ mol/L} \quad (1/2)$$

$$m(\text{solute}) = 0.0251 \text{ mol} \times 137.3 \times 1000 = 3446.21 \text{ mg} \quad (1/2)$$

$$m(1\text{L solution}) = 1.02 \text{ kg} \quad (1/2)$$

$$\text{ppm} = 3446.23 / 1.02 = 3.38 \times 10^3 \text{ ppm} \quad (1/2)$$

The Arrhenius equation for the ionisation of perchloric acid in water is as follows:



- (e) What would be the expected colour of universal indicator in a solution mixture containing 10.0 mL of 0.105 mol/L perchloric acid (HClO_4) and 10.0 mL of 0.105 mol/L ammonia? Explain using a suitable Lowry-Bronsted equation. (2 marks)

Colour: **yellow or orange** (1)

Equation: $\text{NH}_4^+ + \text{H}_2\text{O} \rightleftharpoons \text{NH}_3 + \text{H}_3\text{O}^+$ (1)

- (f) Would you expect a 1.0 molL^{-1} solution of HClO_4 to be a stronger or weaker electrolyte than a 1.0 molL^{-1} solution of CH_3COOH ? Explain. (2 marks)

HClO_4 is a stronger electrolyte than CH_3COOH due to

HClO_4 can ionize fully whereas CH_3COOH can only partially ionize (1)

$[\text{ions}]$ in $\text{HClO}_4 > [\text{ions}]$ in CH_3COOH i.e. more charge carriers in HClO_4 (1)

Question 41

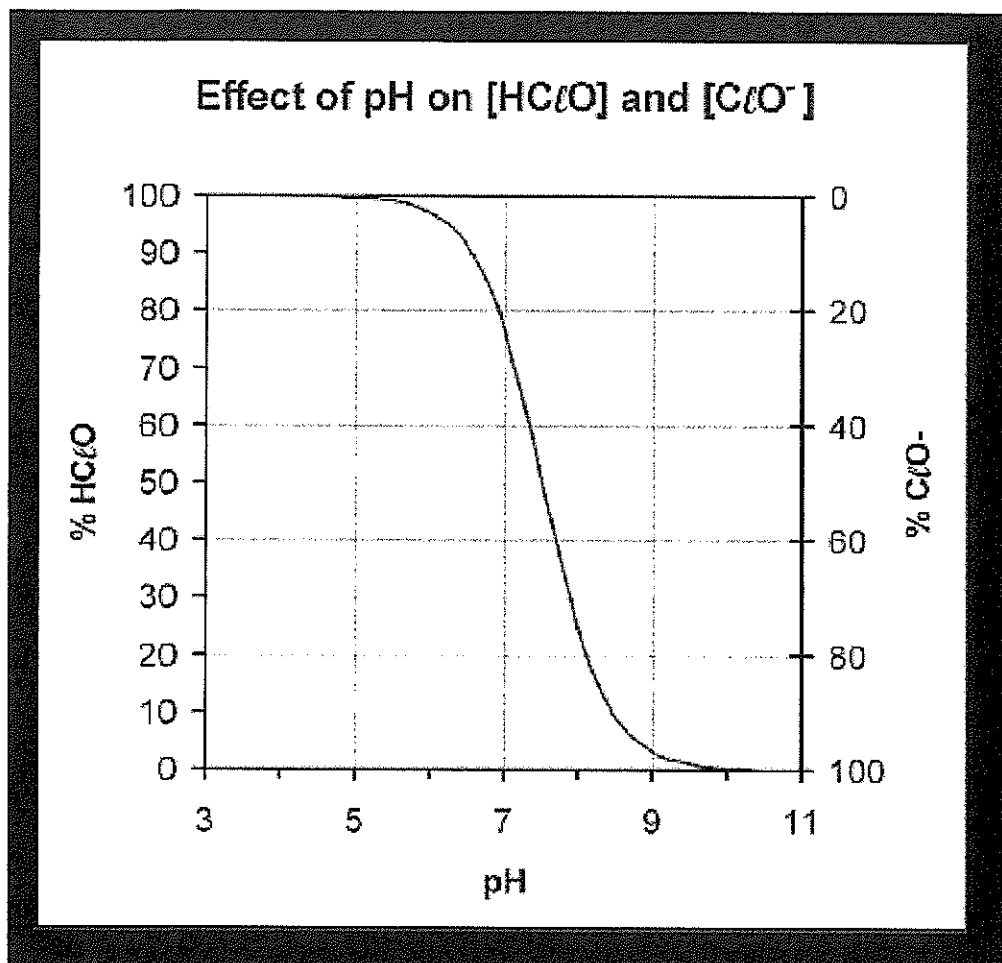
(17 marks)

Common chemicals used to control algae and bacterial growth in swimming pools include hypochlorous acid HClO and the hypochlorite ion ClO^- .

The equilibrium chemical equation between HClO and ClO^- ion in water is



The equilibrium is pH dependent and is very sensitive in the range of pH 7 to 8.



- (a) Examine the graph above and state the effect of pH on the concentration of $\text{HClO}_{(\text{aq})}$ and $\text{ClO}^-_{(\text{aq})}$ ions. (1 mark)

As pH increases, $[\text{HClO}]$ decreases and $[\text{ClO}^-]$ increases. (1/2)

As pH decreases, $[\text{HClO}]$ increases and $[\text{ClO}^-]$ decreases. (1/2)

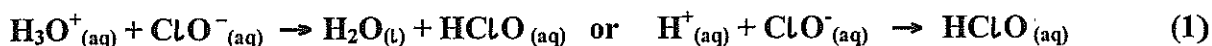
- (b) State the pH at which the concentration of $\text{HClO}_{(\text{aq})}$ equals the concentration of $\text{ClO}^-_{(\text{aq})}$. (1 mark)

7.5 ± 0.1 (1)

- (c) Use Le Chatelier's principle to help explain the shape of the graph. Include in your explanation the effect of low **and** high pH on
- the position of equilibrium and
 - the concentration of HClO and ClO^- .
- (4 marks)

A low pH means an increase in $[\text{H}^+]$ or $[\text{H}_3\text{O}^+]$ in the equilibrium reaction (1/2)

The system counteracts by favouring the reverse reaction where



as a result $[\text{ClO}^-]$ decreases and $[\text{HClO}]$ increases. (1/2)

Two (2) possible answers:

A high pH means a decrease in $[\text{H}^+]$ or $[\text{H}_3\text{O}^+]$ in the equilibrium reaction (1/2)

The system counteracts by favouring the forward reaction where

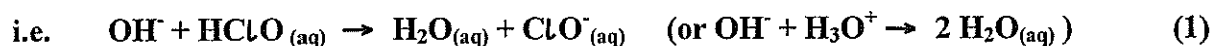


as a result $[\text{HClO}]$ decreases and $[\text{ClO}^-]$ increases. (1/2)

OR ALTERNATIVELY

A high pH means an increase in $[\text{OH}^-]$ in the equilibrium reaction (1/2)

The system counteracts to remove OH^- via the reaction



as a result $[\text{HClO}]$ decreases and $[\text{ClO}^-]$ increases. (1/2)

- (d) Write the equilibrium expression (K) for the reaction: $\text{H}_2\text{O}_{(\text{l})} + \text{HClO}_{(\text{aq})} \rightleftharpoons \text{H}_3\text{O}^+_{(\text{aq})} + \text{ClO}^-_{(\text{aq})}$ (1 mark)

$$K = \frac{[\text{H}_3\text{O}^+] \times [\text{ClO}^-]}{[\text{HClO}]} \quad (1)$$

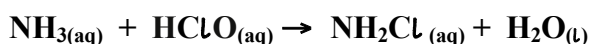
- (e) The equilibrium constant (K) value for the reaction: $\text{H}_2\text{O}_{(\text{l})} + \text{HClO}_{(\text{aq})} \rightleftharpoons \text{H}_3\text{O}^+_{(\text{aq})} + \text{ClO}^-_{(\text{aq})}$ is 3.5×10^{-8} at 25°C . Would you expect the concentration of HClO to be greater, smaller or the same as the concentration of ClO^- ions at 25°C ? Explain your answer. (2 marks)

Small K value means equilibrium reaction favours the reverse pathway (1)

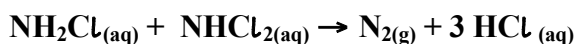
i.e. [reactants] > [products] meaning $[\text{HClO}] > [\text{ClO}^-]$ (1)

Ammonia compounds and organic matter enter a swimming pool from many sources. Swimmers and bathers are major contributors with their bodies giving off saliva, sweat, urine and faecal matter.

The removal of ammonia via chemical reaction with HClO to form chloramine (NH_2Cl) is thus very important:



Chloramine is mildly toxic to bacteria, but tends to be irritating to eyes and skin, so it is an unwelcome addition to the pool. Fortunately, chloramine reacts with each other, as



- (f) An Olympic swimming pool was analysed for ammonia in the water and found to have an ammonia level of 0.175 mg/L . If the swimming pool contains $2,500,000$ litres of water then determine the number of moles of hydrochloric acid produced from the reactions above. (2 marks)

$$m(\text{NH}_3) = (0.175/1000) \times 2500000 = 437.5 \text{ g} \quad (1/2)$$

$$n(\text{NH}_3) = 437.5 / 17.034 = 25.7 \text{ mol} \quad (1/2)$$

$$n(\text{HCl}) = 3/1 \times n(\text{NH}_3) = 77.1 \text{ mol} \quad (1)$$

- (g) Hydrochloric acid lowers the pH of the water in swimming pools. Water with low pH tends to dissolve plaster and grout in swimming pools and corrode metal products.

Sodium hydrogencarbonate (NaHCO_3) or sodium carbonate (Na_2CO_3) powder is often used to remove the hydrochloric acid that lowers the pH of swimming pools and spas.

The **unbalanced** equation for the reaction between sodium carbonate and hydrochloric acid is:



In a swimming pool water testing procedure, a technician recorded a pH reading of 6.50 for a swimming pool containing 2,500,000 litres of water. In order to return the water back to neutral (i.e. a pH reading of exactly 7), he added 44.1 g of 95.0% pure sodium carbonate (Na_2CO_3) to the swimming pool.

Carry out necessary calculations to determine whether the added amount of Na_2CO_3 was able to achieve the desired result (i.e. a final water pH of 7). (6 marks)

$$[\text{H}^+] = 10^{-6.5} = 3.162 \times 10^{-7} \text{ mol/L} \quad (1)$$

$$n(\text{H}^+) = 3.162 \times 10^{-7} \times 2500000 = 0.790 \text{ mol} \quad (1)$$

$$n(\text{Na}_2\text{CO}_3) = 0.95 \times 44.1 / 105.99 = 0.395 \text{ mol} \quad (1)$$

for complete neutralisation, $n(\text{H}^+) = 2 \times n(\text{Na}_2\text{CO}_3)$

$$\text{comparing the number of moles - } 0.790 : 0.395 = 1 : 2 \quad (1)$$

i.e. the added amount of Na_2CO_3 is sufficient to remove all H^+ ions (1)

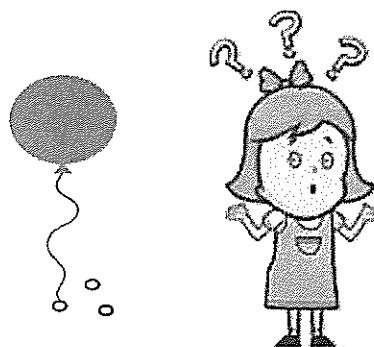
Question 42

(7 marks)

Christine buys some balloons to fill with helium (He) for a party. To stop them floating away, she decides to tie small metal rings to each balloon.

The following information is relevant to the questions that follow.

<i>Mass of a balloon</i>	<i>= 3.60 g</i>
<i>Mass of string tied to each balloon</i>	<i>= 1.50 g</i>
<i>Mass of each metal ring</i>	<i>= 1.20 g</i>
<i>Volume of balloon when inflated</i>	<i>= 15.0 L</i>
<i>Pressure of gas in inflated balloon:</i>	<i>= 103 kPa</i>
<i>Temperature</i>	<i>= 25.0° C</i>



- (a) Determine the maximum number of moles of gas that can be held by the balloon at the conditions stated above. (1 mark)

$$n \text{ (any gas)} = PV / RT \qquad = 103 \times 15.0 / 8.315 \times 298 \qquad = 0.623 \text{ mol} \qquad (1)$$

- (b) Calculate the **minimum** number of metal rings she will need to tie to each balloon to stop it floating away.

Useful information:

i. The balloon will float upwards if the total mass of the balloon, string, metal rings and the helium gas inside the balloon is less than the mass of air (and nothing else!) removed from a fully inflated balloon.

ii. One (1) mole of air has a mass of 28.8 grams. (6 marks)

$$m \text{ (air) in a fully inflated balloon} \qquad = 0.623 \times 28.8 \qquad = 17.957 \text{ g} \qquad (1)$$

$$m \text{ (He) in a fully inflated balloon} \qquad = 0.623 \times 4 \qquad = 2.494 \text{ g} \qquad (1)$$

$$+ \text{ mass of string} \qquad \qquad \qquad + 1.50 \text{ g}$$

$$+ \text{ mass of balloon} \qquad \qquad \qquad + 3.60 \text{ g}$$

$$\text{Total} \qquad \qquad \qquad = \qquad \qquad \qquad 7.594 \text{ g} \qquad (1)$$

$$\text{Extra mass required to keep balloon down} = 17.957 \text{ g} - 7.594 \text{ g} \qquad = 10.363 \text{ g} \qquad (1)$$

$$\text{Number of rings needed} \qquad \qquad \qquad = 10.363 \text{ g} / 1.20 \text{ g} \qquad = 8.62 \qquad (1)$$

$$\qquad \qquad \qquad \sim 9 \text{ rings} \qquad (1)$$